A Survey of AI-Based Methods for SiC MOSFET Management in Automotive Traction Inverters

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***Abstract*—Silicon Carbide (SiC) Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) have become the device of choice for electric vehicle traction inverters, offering higher switching frequencies, lower conduction losses, and superior thermal performance compared to silicon Insulated Gate Bipolar Transistors (Si-IGBTs). However, the increased power density and highly dynamic thermal loading imposed by real drive cycles make junction temperature (Tj) management a critical unsolved reliability challenge. Direct Tj measurement inside packaged modules is impractical, while classical electro-thermal models cannot account for the nonlinear dependence of thermo-sensitive electrical parameters on load current, gate voltage, and device aging simultaneously. This paper surveys the state of the art in artificial intelligence (AI)-based methods addressing three core problems: real-time Tj estimation, aging and remaining useful life (RUL) prediction, and thermally-aware closed-loop control. Neural network surrogates, digital twins with NTC synchronization, condition monitoring via body diode voltage, and equivalent gate resistance controllers are reviewed. Results demonstrate Tj estimation errors below 3.5%, independent tracking of gate oxide and packaging degradation, and lifetime extensions exceeding two times under dynamic load conditions. The survey identifies the absence of a unified, aging-aware, real-time deployable framework as the central open research gap.**

Keywords—SiC MOSFET; traction inverter; junction temperature; artificial intelligence; neural network; digital twin; condition monitoring; remaining useful life; active thermal control; electric vehicles

I. INTRODUCTION

The electrification of transport has placed unprecedented demands on power electronics. The traction inverter, which converts DC battery power into AC current for the electric motor, is the critical component defining the efficiency and reliability of the entire drivetrain. Historically dominated by Si-IGBTs, traction inverters are now transitioning to Silicon Carbide (SiC) MOSFETs. SiC's wider bandgap (~3.26 eV compared to ~1.1 eV for silicon) enables a higher breakdown field strength, faster switching speeds, and lower on-state resistance, which together translate into measurable performance gains: SiC inverters achieve peak efficiencies above 98%, reduce inverter losses by up to 39.8% compared to Si-IGBT equivalents, and extend traction inverter service life by more than 80% in typical driving cycle tests [1]. These achievements cannot be matched by silicon-based semiconductors, particularly on the 800V platforms that now define the direction of modern EV development [10].

However, this performance comes with significant challenges. The higher power density and compact packaging of SiC modules create severe thermal stress under the dynamic, repetitive load cycles of real driving conditions [4]. Three problems stand at the core of reliable SiC MOSFET deployment in traction inverters: junction temperature (Tj) estimation, aging and remaining useful life (RUL) prediction, and thermally-aware real-time control. Conventional approaches to these problems have proven insufficient. Direct temperature sensing requires invasive access to the semiconductor die [2], high-fidelity physics-based thermal models are too computationally demanding for real-time deployment [5], and hardware redundancy adds cost that compounds SiC's already significant price premium over Si-IGBT solutions [3].

Recent advances in artificial intelligence offer a practical path forward. Neural network-based approaches, trained on high-fidelity simulation data, can estimate junction temperatures and detect device degradation in real time with errors below 3.5%, at a fraction of the computational cost of traditional modelling methods [4]. This paper reviews how AI-based methods are being applied to address these three core management challenges of SiC MOSFETs in automotive traction inverters, and examines the extent to which they represent a viable solution for the next generation of EV powertrains.

II. JUNCTION TEMPERATURE ESTIMATION

A. Problem Statement

Inside a traction inverter, a minimum of 6 SiC MOSFETs switch tens of thousands of times per second. Each switching event deposits a pulse of heat directly into the semiconductor junction. Junction temperature (Tj) refers to the operating temperature at the semiconductor die itself, located within the power module package. Thermal factors account for approximately 55% of all power electronics failures [4]. The resulting repetitive thermal cycling due to acceleration and regenerative braking under real driving conditions leads to thermal swings within Tj, producing thermomechanical stress across the material layers of the power module, accelerating degradation mechanisms such as bond wire lift-off and solder fatigue, and ultimately leading to device failure [1]. Accurate knowledge of Tj is therefore essential for preventing failure of SiC MOSFETs, as well as state-of-health monitoring and RUL prediction.

The fundamental challenge is that Tj cannot be measured directly during operation. Placing a physical sensor inside the module would require removal of the dielectric gel or structural modification of the package, rendering the device inadequate for industrial deployment [2].

B. Conventional Approaches and Limitations

Electro-thermal modelling calculates Tj by measuring heatsink temperature and combining it with estimated device losses and a fixed lumped thermal model. These models are defined under constant operating conditions and cannot incorporate aging effects, meaning their accuracy degrades progressively over the device lifetime [2].

NTC thermistors placed near the chip offer a simple hardware solution, but face an irresolvable placement dilemma: positioning them close enough to the SiC die for thermal accuracy exposes them to insulation breakdown risk from high switching voltages, while placing them on a separate PCB for electrical safety introduces sufficient thermal distance to make the measurements unreliable.

Temperature Sensitive Electrical Parameters (TSEPs), particularly Rds,on and Vgs,th, offer a non-intrusive alternative by exploiting the temperature dependence of electrical properties already measurable at the gate driver [9]. However, Rds,on depends simultaneously on temperature, load current, gate voltage, and aging state, creating a nonlinear mapping that fixed lookup tables and formulas cannot reliably resolve across all operating conditions, particularly in the electrically noisy, high-frequency switching environment of SiC inverters [2].

The 4H-SiC PN diode integrated sensor represents another approach, demonstrating stable operation from 20°C to 600°C with a sensitivity of 2.8 mV/K at 1 μA forward current [7]. However, this sensitivity is extremely small relative to the voltage transients during switching, which can reach hundreds of volts per microsecond. Furthermore, the reverse biased sensing mode becomes unreliable below 250°C, covering only the upper portion of the traction inverter operating range [7].

C. AI-Based Solutions

The fundamental limitation shared by all conventional approaches is their inability to handle the nonlinear, multi-variable, time-varying relationship between measurable signals and junction temperature. This is precisely where AI-based methods offer a decisive advantage.

Egin Martin et al. [4] addressed this by building a high-fidelity CFD model of a dual SiC traction inverter, then developing a Reduced Order Model (ROM) to efficiently generate training data, before training an ANN on over 130 combined CFD and ROM results. Taking coolant flow rate, inlet temperature, and heat loss as inputs, the trained ANN estimates maximum junction temperature in approximately one second, reducing computation time from 150 minutes while maintaining prediction errors below 3.5%. The model was experimentally validated on a hardware-in-the-loop platform under real drive cycle conditions, demonstrating robust real-time performance.

Troldborg et al. [5] proposed a neural network-based digital twin targeting the full three-dimensional temperature distribution inside the power module rather than a single maximum value. A FEM model built in COMSOL Multiphysics generates a labelled dataset across a parametric sweep of operating conditions, on which a neural network with 7 hidden layers and 128 neurons per layer is trained, achieving estimation of all ten individual junction temperatures of a Wolfspeed SiC module in 0.063 seconds. Critically, NTC thermistor feedback is incorporated to maintain synchronization between the digital model and the physical module as it ages. A double-NTC variant, placing thermistors above and below the DCB solder layer, allows the digital twin to detect which specific material layer is degrading, extending the solution beyond temperature estimation into early degradation detection [5].

Together these two approaches demonstrate that AI-based methods can provide accurate, non-intrusive, real-time Tj estimation without any physical access to the semiconductor die, resolving the core limitations of all conventional approaches simultaneously.

III. AGING AND REMAINING USEFUL LIFE PREDICTION

A. Problem Statement

SiC MOSFETs in traction inverters degrade over their operational lifetime through two distinct categories of failure mechanisms. At the chip level, gate oxide degradation driven by high electric field stress and bias-temperature instability (BTI) causes threshold voltage drift, increasing leakage current and ultimately leading to gate oxide breakdown. At the package level, the coefficient of thermal expansion (CTE) mismatch between the SiC die and surrounding materials causes thermomechanical stress under repeated thermal cycling, leading to bond wire heel cracking, bond wire lift-off, and die attach solder fatigue [3]. Because SiC chips are significantly smaller than equivalent Si devices at the same power level, current density is higher, making them more susceptible to package-level degradation [3].

The core challenge is that these two degradation modes occur simultaneously and affect the same measurable electrical parameters in different ways. Rds,on increases due to both gate oxide degradation and bond wire fatigue, meaning a single parameter cannot distinguish between the two failure modes. Vth tracks gate oxide degradation but is insensitive to packaging failures. No single conventional precursor parameter provides a complete and independent picture of device health [3].

B. Conventional Approaches and Limitations

Physics of Failure (PoF) models, such as the Bayerer (CIPS 2008) model and the LESIT lifetime model, estimate the number of cycles to failure as a function of junction temperature swing and mean temperature [6]. These models are useful for offline lifetime estimation but require accurate knowledge of Tj history across the full operating life, which as established in Section II, is itself difficult to obtain. Furthermore, PoF models are defined under controlled accelerated test conditions and do not generalize well to the irregular, highly dynamic load profiles of real automotive driving cycles. Rainflow counting techniques are used to extract thermal cycles from real drive cycle data, but the accuracy of the resulting RUL estimate depends on how well model parameters match actual device behavior, which degrades as the device ages [6].

C. AI-Based Solution

The condition monitoring approach proposed by Ugur et al. [3] addresses the independent tracking problem directly by exploiting a unique property of SiC MOSFETs in third quadrant operation. When the device conducts through its MOS channel at negative gate bias, the body diode voltage drop (VSD) reflects the condition of the MOS channel and is sensitive to gate oxide degradation. When the device conducts through its PiN diode path, VSD reflects only contact resistance variations, making it sensitive exclusively to packaging degradation such as bond wire cracking and solder fatigue. By monitoring VSD at two gate bias levels (0V and -5V), gate oxide and packaging degradation can be tracked independently using a single measurable parameter, without requiring any additional sensors or package modifications [3]. Experimental validation on artificially degraded devices confirmed the method's ability to independently identify both degradation modes.

For RUL prediction, the Chalmers thesis [6] demonstrates that data-driven models and PoF models work best in combination. PoF models provide the physical degradation framework, while machine learning supplies the ability to adapt that framework to real-world conditions using continuously updated measurements. The key advantage of integrating AI into this pipeline is that as the device ages and its electrical parameters drift, the AI model can update its internal state using tracked precursor parameters rather than relying on fixed datasheet values, continuously refining the RUL estimate throughout the device lifetime.

The remaining open challenge is that SiC is a relatively new technology and long-term field data is limited. Most AI models for RUL prediction have been trained on accelerated aging test data rather than real driving cycle degradation histories, and their generalizability to actual automotive deployment conditions remains an active research question.

IV. THERMALLY-AWARE REAL-TIME CONTROL

A. Problem Statement

Knowing Tj in real time is only useful if the inverter controller can act on that information to prevent damage and extend device lifetime. Without closed-loop thermal control, the inverter operates on fixed switching parameters regardless of the actual thermal state of the SiC MOSFETs. Under real driving conditions with dynamic load changes, this means the device either operates conservatively below its true thermal limit, wasting performance headroom, or experiences uncontrolled Tj swings that accelerate degradation. The fundamental objective of thermally-aware control is to use real-time Tj knowledge to actively regulate the temperature swing, reducing thermomechanical stress and extending lifetime without sacrificing efficiency unnecessarily.

B. Conventional Approaches and Limitations

Switching frequency control dynamically adjusts the PWM frequency in response to Tj feedback: at high junction temperature the frequency is reduced to cut switching losses, and at low junction temperature it is increased to recover efficiency. While conceptually simple, this approach trades efficiency for thermal protection reactively rather than predictively, and cannot distinguish between a transient thermal spike and a sustained overload [6]. Gate resistance manipulation, implemented through adaptive gate drivers, controls the speed of switching transitions by varying the gate charging resistance, adjusting switching losses in real time. This provides finer control than frequency adjustment alone but adds hardware complexity [6]. Both strategies suffer from the same fundamental limitation: they react to temperature after it has already risen, rather than anticipating and preventing dangerous thermal conditions.

C. AI-Based Solution

The Equivalent Gate Resistance Control (EGRC) method proposed in [8] directly addresses this by using real-time Tj measurement as the feedback signal for an active thermal management controller. The system continuously monitors Tj at each SiC MOSFET, computes the deviation from a reference temperature, and dynamically adjusts the gate resistance through an auxiliary switch circuit to modify switching losses in real time. Experimental results demonstrated that this approach reduced the maximum junction temperature fluctuation range from 18.83T to 9.85°C under power fluctuation conditions, prolonging the estimated SiC MOSFET lifetime by 2.18 times. The paper also introduces the concept of a Temperature Control Operating Space (TCOS) and a Junction Temperature Control Coefficient (JTCC) to bound the control action, preventing the controller from over-protecting the device at the cost of system performance [8].

The Chalmers thesis [6] demonstrates a PWM frequency-based hysteresis controller that uses Tj as the control variable, switching between maximum and minimum switching frequencies based on defined thermal thresholds. When combined with accurate AI-based Tj estimation rather than slow NTC feedback, this type of controller gains a decisive advantage: it can act on the true junction temperature rather than a lagged approximation, enabling tighter thermal control with smaller protective margins and therefore higher average performance.

The most complete vision of AI-driven thermal control is the digital twin predictive control framework described by Troldborg et al. [5], where the neural network digital twin predicts future Tj based on current operating conditions before the temperature actually rises, allowing the controller to take preventive action rather than reactive action. For a predictive control strategy to be effective, the computation time of the thermal model must be significantly shorter than the thermal time constant of the system. The 0.063 second inference time of the neural network digital twin satisfies this requirement, making genuinely predictive closed-loop thermal control feasible for the first time in this application context.

V. CONCLUSION

This paper has reviewed the application of artificial intelligence methods to three core challenges in the management of SiC MOSFETs in automotive traction inverters: junction temperature estimation, aging and remaining useful life prediction, and thermally-aware real-time control.

Across all three problems, a consistent pattern emerges. Conventional approaches, whether physics-based thermal models, direct sensing methods, or fixed control strategies, are individually insufficient because they cannot handle the nonlinear, multi-variable, and time-varying nature of SiC MOSFET behavior under real driving conditions. AI-based methods, particularly artificial neural networks and digital twins, address this shared limitation by learning complex mappings directly from high-fidelity simulation data and adapting to real operating conditions through sensor feedback.

The results reported in the reviewed literature are promising. ANN surrogates trained on CFD and FEM data achieve junction temperature estimation errors below 3.5% with inference times under one second, making real-time deployment feasible. Condition monitoring approaches exploiting the unique third-quadrant conduction behavior of SiC MOSFETs enable independent tracking of gate oxide and packaging degradation from a single measurable parameter. Active thermal management controllers using real-time Tj feedback have demonstrated lifetime extensions of over two times under dynamic load conditions.

However, significant open challenges remain. Most AI models have been validated under controlled laboratory or simulation conditions rather than real automotive drive cycles. The generalizability of these models to aging devices operating across the full range of EV operating conditions remains an active research question. Furthermore, no unified framework currently integrates all three functions — Tj estimation, RUL prediction, and thermally-aware control — into a single deployable system. Addressing this integration gap represents the most critical direction for future work in this field.

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